

TERM PAPER

POLYMER MEMBRANES FOR FUEL CELL



CHEM-535 (POLYMER SYNTHESIS)

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ABSTRACT

Fuel Cells (FCs) are an ecofriendly device, generates electricity from chemical energy where polymer membranes play a pivotal role in ion transport and system efficiency. According to electrolyte type, various types of fuel cells such as Proton exchange membrane fuel cell (PEMFC), Anion Exchange Membrane fuel cell (AEMFCs), and Direct methanol fuel cell (DMFC), are used both in laboratory and industrial applications. Polymer membranes, the heart of the FCs are primarily classified as– Perfluorosulfonic acid (PFSA) and hydrocarbon (HC) based polymers. PFSA, such as Nafion, has been considered a standard membrane, due to their high proton conductivity and excellent durability. However, poor methanol resistance and high cost limit its viability. To tackle this, aromatic based membranes emerge as a promising alternative since the backbone of these polymers are linked with aryl rings, ether, ketones, sulfones, and benzimidazole linkages that provide excellent thermal and mechanical strength, smaller methanol permeabilities, and most significantly lower costs. Remarkably, the electrochemical characteristics and durability of these PEMs are possible to enhance by structural modification, crosslinking, polymer blending and so on. Therefore, HC based polymers hold abundant potential to replace the Nafion, yet there are still few obstacles since crosslinking may lead to brittleness and fillers could form circuitous pathway for proton conductivity. Nonetheless, the well balance among all the individual characteristics determines the success of these PEMs. This term paper presents a comprehensive overview of different types of PEMs, including all the relevant characteristics, proton transportation mechanisms, synthesis methods, and the major challenges involved in developing next-generation fuel cells.

2. INTRODUCTION:

The production of energy through renewable sources has shown a global increase in recent decades. The non-renewable resources usually include burning fossil fuels or nuclear energy which is the prime reason for several health problems and environmental dangers. Alternatively, renewable sources such as geothermal, wind, solar, fuel cells are ecofriendly resources both in laboratories and industrial scale. Among these renewable sources, fuel cells gain widespread use due to their unique properties as exemplified by their high efficiency, low emission, quick startup and desirable power density. According to the 2017 report by the U.S. Department of Energy, global fuel cell power shipments exceeded 650 MW, and the transportation sector demonstrating the most significant increase in installed capacity.

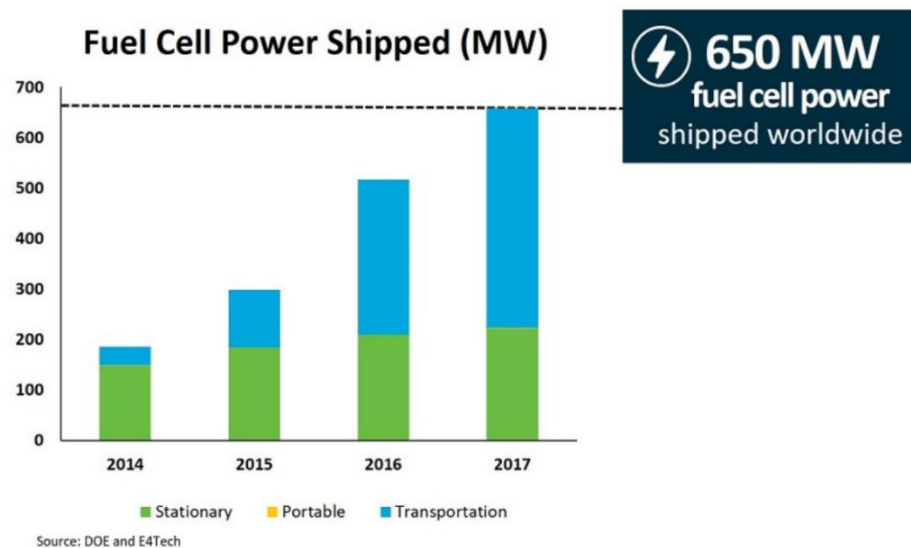


Fig-1: Global FCs power shipment (2014-2019)

2.1 Fuel Cells and its operation:

A fuel cell is an energy conversion device that generates electricity and heat by electrochemically combining a gaseous fuel (hydrogen) and an oxidant gas (oxygen). The first fuel cells were invented in 1838 by Sir William Grove. Unlike a battery, a fuel cell does not require any recharging rather it produces energy as long as fuel is provided. The cell is mainly composed with of two electrodes—

anode and cathode, catalysts and an electrolyte that allow the ions to move between the two side of the cells. Hydrogen fuel is fed into the electrode anode while oxygen gas enters through the cathode. Catalytic action happens simultaneously both at anode and cathode. As a result, hydrogen splits into protons and electrons; the protons pass through the electrolyte, while the electrons generate electricity before combining oxygen to form heat and water as byproduct.

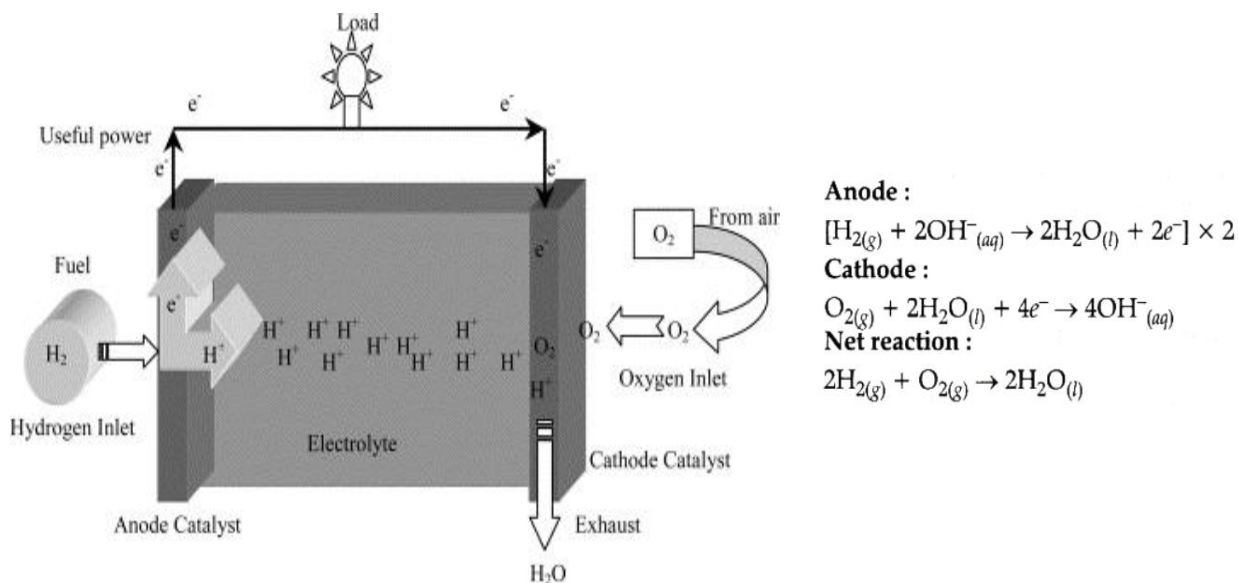


Fig:2 Structure of a fuel cell

3.2 Types of fuel cells:

The classification of fuel cells is primarily based on the type of electrolyte used and their operating temperature. There are numerous types of fuel cell technologies being developed for different applications, some of which are summarized in **Table 1**

Table-1: Types of FCs and their applications						
Types	Anode	Electrolyte	Cathode	Operating T (C)	Fuel	Application
DMFC	Pt-Ru	Nafion polymer electrolyte	Pt	60–200	Liquid methanol	Portable power
PEMFC	Pt	Nafion polymer electrolyte	Pt	50–80	Less pure hydrogen from hydrocarbons or methanol	Automobile

AFC	H ₂	KOH	O ₂	50–200	Pure hydrogen, or hydrazine	Areospace
PAFC	H ₂	H ₃ PO ₄	O ₂	160-210	Hydrogen from hydrocarbons and alcohol	Stationery power generation
SOFC	H ₂	Ceramic composed of ZrO ₂	O ₂	600–1000	Natural gas or propane	Stationery power generation

3.3 Polymer Membranes in fuel cells:

A solid ion-conducting polymer electrolyte membrane (PEM) is one of the vital core components in fuel cell systems such as Proton Exchange Membrane Fuel Cell (PEMC), Anion Exchange Membrane Fuel Cell (AEMC). The membrane primarily works as a separator between the anode and cathode, repels electrons as well as acts as a barrier between the fuel and oxidants. The electrodes, catalyst, and the membrane form the membrane electrode assembly. The Hydrogen fuel and air are supplied on either side through channels formed in flow field plates. PEMCs are cationic exchange membranes possess negative charged groups e.g. SO_3^- , whereas AEMC holds positive functional charged e.g. $-\text{NH}_2^+$ on the membrane's

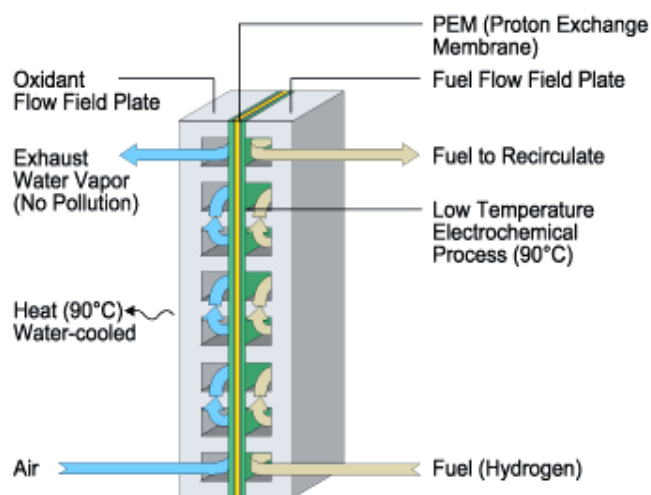


Fig:3 Structure of a PEM

polymeric backbone. The durability and performance in fuel cells requires that these ion exchange membranes have high proton/anion conductivity, good resistance to fuel crossover, excellent mechanical, thermal, and chemical strength. PFSA based polymer, such as Nafion, has been considered a standard membrane, due to their high proton conductivity and excellent durability. However, poor methanol resistance and high cost limit its viability. To tackle this, aromatic based polymer membranes emerge as a promising alternative since the backbone of these polymers are linked with aryl rings, ether, ketones, sulfones, and benzimidazole linkages

that provide excellent thermal and mechanical strength, smaller methanol permeabilities, and most significantly lower costs.

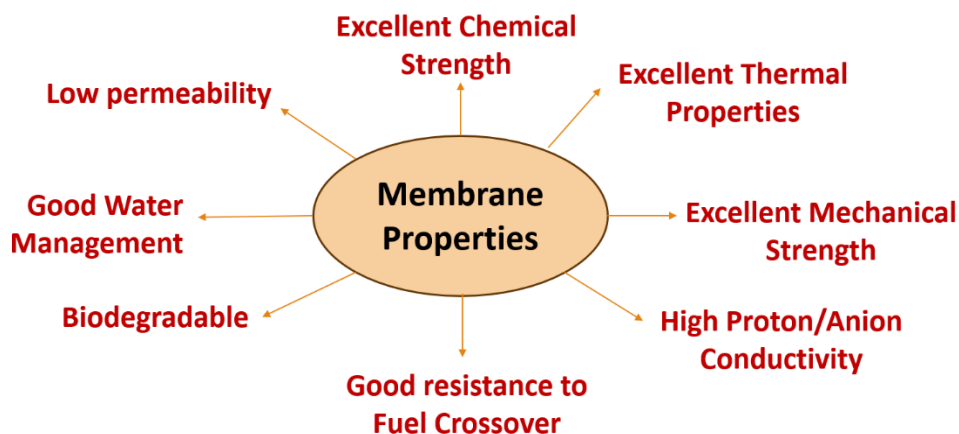


Fig:4 Key properties of PEM

2.4 Proton Conduction Mechanisms in PEM: Proton conduction in PEMs fundamentally occurs via two intertwined mechanisms– Grotthuss mechanism, and the Vehicle mechanism.

- **Grotthuss mechanism:** In 1806, Theodor Grotthuss proposed this theory. This mechanism commonly known as Hopping mechanism. The proton transfer takes place between oxygen atoms through hydrogen bond formation and cleavage and requires the presence of an uninterrupted hydrogen-bond network in the proposed materials. In PEMs, Grotthuss mechanism is influenced by the membrane's hydrophilic clusters, temperature, and water content. Therefore, understanding this mechanism assists in designing better PEMs, such as those with optimized hydrophilic channels and functional groups, to enhance proton conductivity under various operating conditions, even at lower humidity.

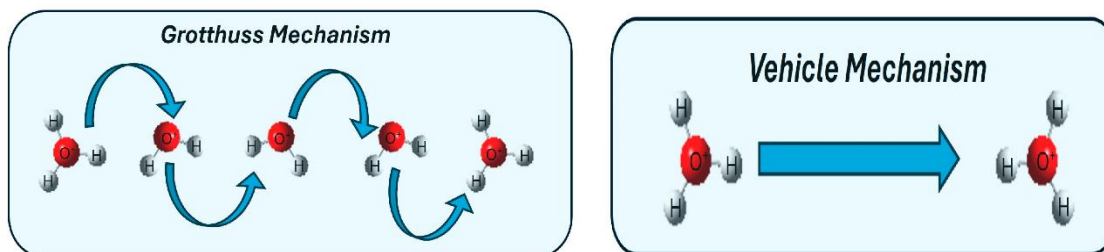


Fig:5 Proton conduction mechanism in PEM

- **Vehicle mechanism:** This mechanism was Proposed by Kreuer, Rabenau, and Weppner in 1982. It is commonly known as Ion diffusion mechanism where proton transfer occurs bt diffusion of the proton. The water molecules (H_2O) act as vehicles to transport the proton from the anode to cathode electrode by the formation of H_3O^+ . During vehicle operation, the fuel cell faces varying loads, that impacts ion transport. Thus, it makes these mechanisms vital for modeling and improving longevity of a FC.

2.5 Types of polymer membranes:

Polymer membranes are primarily classified as—Perflorosulfonic acid (PFSA) and Hydrocarbon (HC) based polymers. PFSA, such as Nafion, has been regarded as a standard membrane, due to their high proton conductivity and excellent durability. However, this membrane is extremely costly as well as has poor methanol resistance that limits its viability. To tackle this, HC based membranes emerge as a promising alternative. The polymeric backbone of these polymers is linked with aryl rings, ether, ketones, sulfones, and benzimidazole linkages. The aromatic rings provide rigidity, which leads to thermal and mechanical stability. As a result, a variety of hydrocarbon-based polymers, including poly-ether ketones (PEK) Poly vinyl alcohol (PVA), poly (ether ether ketone) (PEEK), sPEEK, PBI, polyesters, and polyimides (PIs), have been actively researched and developed for use in FC applications. Each of these has own advantages and disadvantages. Some of these membranes along with Nafion are summarized in **Table-2**

Table-2 Different kinds of PEMs		
Polymer electrolyte membrane	Advantages	Disadvantages
Nafion based	• High mechanical stability. • Can be performed in a thin-film structure	• High cost production • High ethanol/methanol permeability
Sulfonated poly ether ether ketone	• Excellent mechanical and thermal stability. • Good compatibility.	• Difficult to control the modification.
Poly (vinyl alcohol)	• Easy to modify chemically and physically. • Low-cost and biodegradable	• High swelling ratio.
Polybenzimidazoles based membrane	• Good electrochemical properties. • Low membrane resistances. • High thermal stability	• Low mechanical stability
Chitosan based membrane	• Biopolymer materials • Low-cost and biodegradable	• Low ionic conductivity • High water uptake due to high hydrophilic properties

3.Synthetic Route for Polymer Membrane: The three-reaction schemes for PEM are outlined below—

3.1 Nafion Monomers for PFSA: Nafion is a widely used membrane in Perfluorosulfonic acid polymer (PFSA). The structural morphology of Nafion affects the mechanical and thermal properties of the polymer backbone. Thus, it is important to analysis the morphological model structure to clarify the unique properties of Nafion. The first model for Nafion—known as the **cluster-channel or cluster-network model**—describes the material as containing evenly distributed sulfonate-rich clusters,

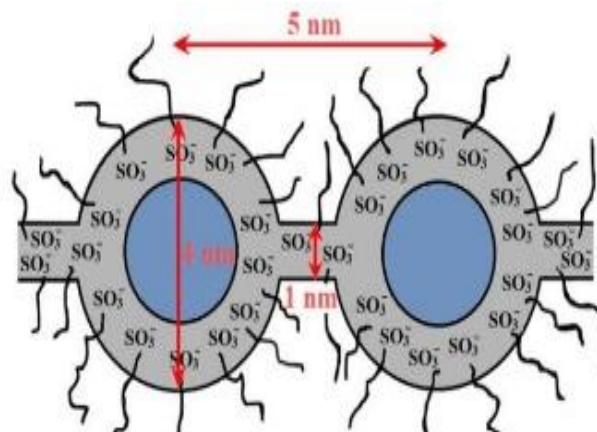


Fig-6 Cluster-network model of Nafion

sometimes referred to as *inverted micelles*. These clusters are roughly **40 Å (4 nm)** in diameter and are embedded throughout a continuous fluorocarbon matrix. They are connected by narrow **~10 Å (1 nm)** wide channels, which account for Nafion’s characteristic ion-transport behavior.

- **Key functionality:** The **sulfonic acid groups** located at the side chain of the polymeric backbone perform as a hydrophilic region. It absorbs water and generates “ion-clusters” between the hydrophobic matrixes. These channels facilitate the rapid transport of cations via **Vehicle and Grotthuss mechanisms**, making it an excellent PEM in fuel cells.
- **Synthesis Scheme:** Nafion, is synthesized by free-radical copolymerization of perfluorinated vinyl ether (PFVE) co-polymer with tetrafluoroethylene (Teflon). The resulting copolymer contains $\text{—SO}_2\text{F}$ groups, which are hydrolyzed with hot aqueous NaOH to form $\text{—SO}_3^-\text{Na}^+$. This is then acidified to give the $\text{—SO}_3\text{H}$ form of Nafion. The mechanism of this membrane is outlined below—

concentration, and sulfonation time. SPEEK demonstrated desirable chemical durability at low DS, with greater dimensional, mechanical, and thermal stability than Nafion but lower water uptake and proton conductivity.

- **Synthesis Scheme:** A specific sulfonated dihalogenated diaryl ketone monomer or sulfonated bisphenol derivative can be directly copolymerized with a suitable dihalogenated diaryl ether monomer unit to produce random (statistical) sPEEK.

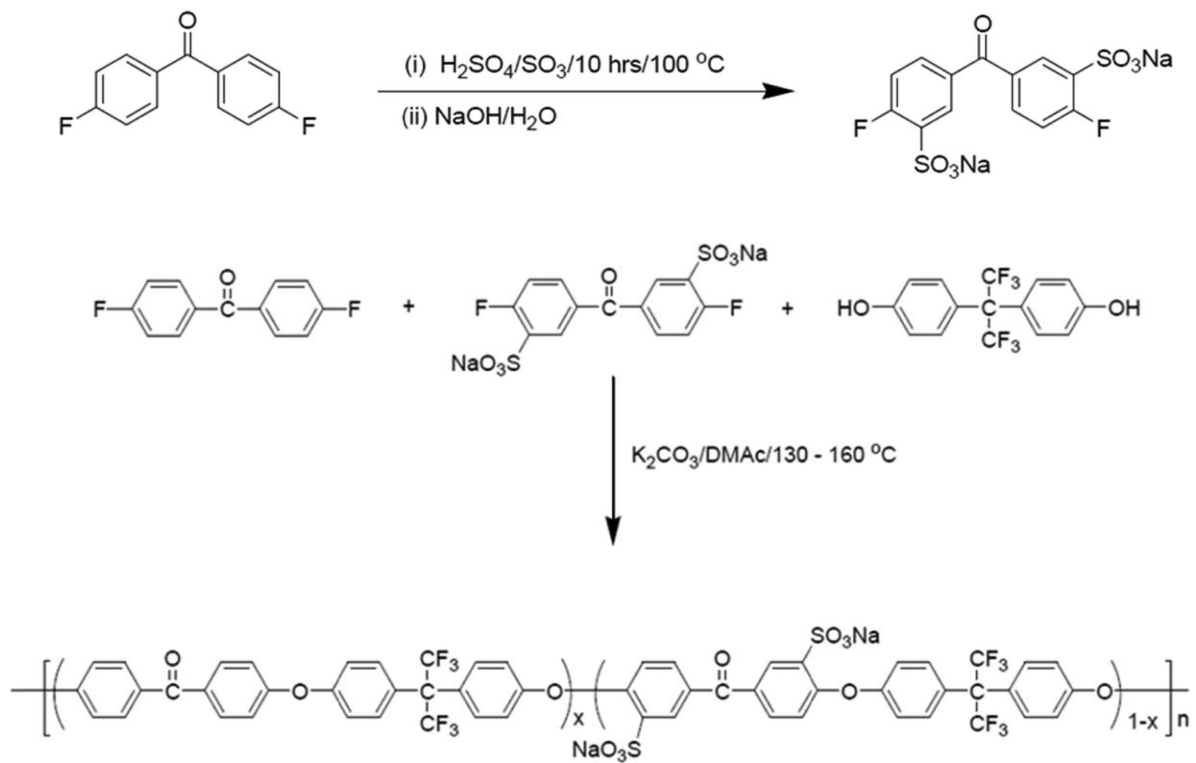


Fig-8 (A) Reaction mechanism for the preparation of sPEEK via base catalyzed

On the other hand, **PEEK** is a highly stable polymer with very low solubility in organic solvents. Adding sulfonic acid groups directly to the PEEK backbone lowers its crystallinity and improves solubility. Therefore, direct sulfonation using agents like sulfuric acid occurs through electrophilic substitution generated sPEEK.



Fig-8 (B) Reaction of sPEEK via Direct post-polymerization sulfonation reactions with PEEK

3.3 Polybenzimidazole (PBI): PBI is well known for its excellent thermal and chemical stability. It is widely used to make membranes for **high-temperature** fuel cells operations. Additionally, this membrane can perform its operations under strongly acidic and harsh solvents environments. The incorporation of inorganic nano-fillers into the PBI matrix further augmented its performance.

- **Synthesis scheme:** The synthesis of PBI membrane involves the polycondensation reaction of aromatic tetraamines– 3,3',4,4'tetraamino-biphenyl (TAB) and dicarboxylic acids, in the presence of polyphosphoric acid (PPA).

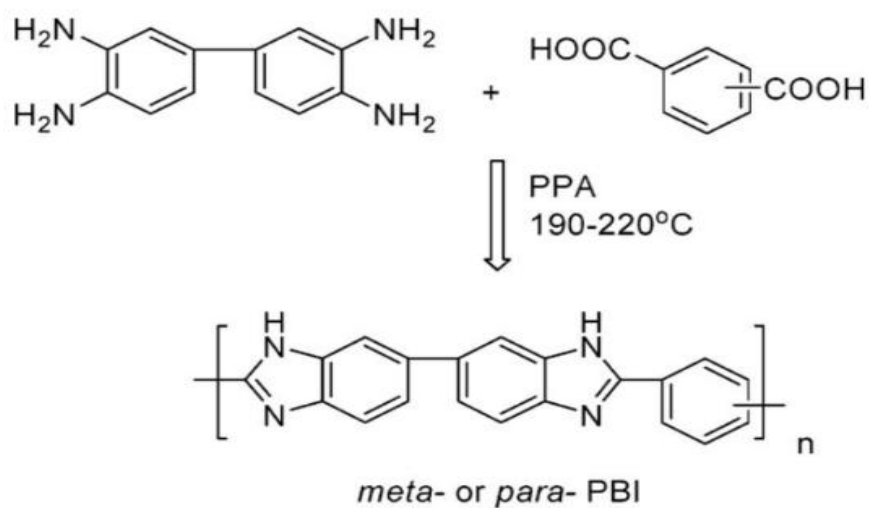


Fig-9: PBI synthesis by condensation reaction

4.Challenges and Future Perspectives:

Nafion has been widely commercialized as a trademark in FC industry. Recent advances in this area have demonstrated that HC-based PEMs have strong potential to serve as solid electrolyte alternatives to Nafion. In addition, several modification strategies have improved their electrochemical performance and durability, but long-term stability under industrial scale is still not fully explored. Although such polymers–PEEK, PBI, PES, PI and PPO are produced on an industrial scale, their proton-conducting versions have not yet been successfully commercialized due to durability and stability limitations. Thus, HC based PEMs made by designed monomers may offer better durability but may face challenges in complex synthesis at large scale. To sum up, functionalization, blending, or filler addition to commercial polymers appears more practical, though optimization of properties, durability, and cost remains essential.

5. Conclusion:

PEMs play a critical role in the performance, efficiency, and durability of FC system. This term paper reviewed different types of fuel cells and highlighted the importance of PEMs as proton-conducting electrolytes. The detailed assessment of Nafion, including its synthesis and structural morphology depicts its high proton conductivity and long-standing role as the commercial benchmark membrane for FC applications. However, high cost and underperformance of Nafion at elevated temperatures, draw attention to developing alternative polymer membranes. Although HC-based membrane emerged as promising alternatives, challenges related to long-term, and large-scale production create this unsettled issue.

Overall, continued research into membrane synthesis, functionalization, and structural optimization is essential for developing next-generation polymer membranes. Advances in these areas will enable more efficient, durable, and cost-effective fuel cells, supporting their broader adoption in clean energy technologies.

Q & A

I. What are the major modification strategies used to improve the structural and functional performance of PEM?

Crosslinking, multiblock-copolymerization, the introduction of inorganic or organic fillers, nano-fillers, the addition of branching or pendant structures, and morphological modifications within the polymer matrix have been proven to improve the mechanical, thermal, oxidative stability, fuel resistance, and electrochemical performance of PEM.

II. How does temperature affect the proton conductivity of the polymer membrane in fuel cells?

Temperature has a dual effect on proton conductivity. High temperatures may increase evaporation, which potentially dries out the membrane. Similarly, increasing temperatures provide the necessary thermal energy for proton hopping in the Grotthuss mechanism.

III. Why are aromatic-based polymer membranes promising candidates for next-generation PEM fuel cells?

Aromatic-based membranes containing polymer chains are linked together by aromatic rings, alongside varying numbers of ether, ketone, imide, or benzimidazoles in their structures possess smaller methanol permeabilities compared to Nafion. These membranes can also be chemically modified and work well at elevated temperatures, such as PBI.

Abbreviations

FCs	Fuel Cells
HCFC	Hydrocarbon fuel cell
PEM	Polymer Electrolyte Membrane
PEMC	Proton Exchange Membrane Fuel Cell
AEMC	Anion Exchange Membrane Fuel Cell
PFSA	Perfluorosulfonic acid
DMFC	Direct methanol Fuel cell
AFC	Alkaline Fuel cell
PAFC	Phosphorous Acid Fuel cell
SOFC	Solid Oxide Fuel Cell
PTEF	Polytetrafluoroethylene
PFVE	Perfluorinated Vinyl Ether

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